



Mobile Application Development

QP Code: SSC/Q8407

Version: 1.0

NSQF Level: 4.5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
Noida Uttar Pradesh - 201303 || email:sscstandards@nasscom.in

Qualification Pack

Contents

SSC/Q8407: Mobile Application Development	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8437: Work Organization and Management for Mobile Application Development	5
SSC/N8438: Communication and Interpersonal Skills for Mobile Application Development	8
SSC/N8439: Sustainable Practice	11
SSC/N8440: Initial Planning, Design, and Test Framework	14
SSC/N8441: Systems Architecture Planning	17
SSC/N8442: Implementation and Product Development	20
SSC/N8443: Final Product Tests, Troubleshooting, and Optimization	24
Assessment Guidelines and Weightage	26
<i>Assessment Guidelines</i>	26
<i>Assessment Weightage</i>	27
Acronyms	28
Glossary	29

Qualification Pack

SSC/Q8407: Mobile Application Development

Brief Job Description

Individuals at this job are responsible for designing and maintaining mobile-based applications, system architecture planning, developing back-end APIs, product testing, fixing of bugs, etc. They may work stand alone or along with application/functional developers as part of the overall solution that includes a mobile-based component. This job may require the personnel to work as an individual contributor or manage a team of junior contributors.

Personal Attributes

Individuals in this role must work and collaborate with various stakeholders involved in the development of web-based or mobile-based applications and solutions. They must be able to communicate and build relationships with others and continuously develop their knowledge and analytical abilities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8437: Work Organization and Management for Mobile Application Development](#)
2. [SSC/N8438: Communication and Interpersonal Skills for Mobile Application Development](#)
3. [SSC/N8439: Sustainable Practice](#)
4. [SSC/N8440: Initial Planning, Design, and Test Framework](#)
5. [SSC/N8441: Systems Architecture Planning](#)
6. [SSC/N8442: Implementation and Product Development](#)
7. [SSC/N8443: Final Product Tests, Troubleshooting, and Optimization](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
Country	India

Qualification Pack

NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ NIL
Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) OR Pursuing 1st year of UG OR Previous relevant Qualification of NSQF Level (4) with 1-2 Years of experience relevant field
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	28/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-IT-01806-2024-V1-NASSCOM
NQR Version	1.0

Qualification Pack

SSC/N8437: Work Organization and Management for Mobile Application Development

Description

Work Organization and Management for Mobile Application Development

Scope

The scope covers the following :

- Work Organization and Management for Mobile Application Development

Elements and Performance Criteria

Work Organization and Management for Mobile Application Development

To be competent, the user/individual on the job must be able to:

- PC1.** Organize and maintain a safe and efficient workspace
- PC2.** Maintain the integrity and confidentiality of systems, data, information and documents at all times
- PC3.** Acquire, use, maintain, and store all equipment and materials to ensure optimal and sustained performance
- PC4.** Read, appraise and clarify the rights and obligations tied to formal documentation of all kinds
- PC5.** Review opportunities, expectations and offers, relative to personal professional capacity, in order to make open, informed choices
- PC6.** Select, use and keep up to date selected measures for work planning, scheduling and prioritization
- PC7.** Check and ensure that all specific resources are available for the work in hand
- PC8.** Meet or enhance the satisfaction of clients and others through self-knowledge, expectation management, and personal efficiency and effectiveness
- PC9.** Proactively grow personal expertise through research and, professional development

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work Organization and Management for Mobile Application Development</i>	20	35	-	-
PC1. Organize and maintain a safe and efficient workspace	2	4	-	-
PC2. Maintain the integrity and confidentiality of systems, data, information and documents at all times	2	4	-	-
PC3. Acquire, use, maintain, and store all equipment and materials to ensure optimal and sustained performance	3	4	-	-
PC4. Read, appraise and clarify the rights and obligations tied to formal documentation of all kinds	2	4	-	-
PC5. Review opportunities, expectations and offers, relative to personal professional capacity, in order to make open, informed choices	2	4	-	-
PC6. Select, use and keep up to date selected measures for work planning, scheduling and prioritization	2	4	-	-
PC7. Check and ensure that all specific resources are available for the work in hand	2	4	-	-
PC8. Meet or enhance the satisfaction of clients and others through self-knowledge, expectation management, and personal efficiency and effectiveness	3	4	-	-
PC9. Proactively grow personal expertise through research and, professional development	2	3	-	-
NOS Total	20	35	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8437
NOS Name	Work Organization and Management for Mobile Application Development
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8438: Communication and Interpersonal Skills for Mobile Application Development

Description

Communication and Interpersonal Skills for Mobile Application Development

Scope

The scope covers the following :

- Communication and Interpersonal Skills for Mobile Application Development

Elements and Performance Criteria

Communication and Interpersonal Skills for Mobile Application Development

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare for meetings with clients and associates
- PC2.** Gather, clarify, and confirm client requirements
- PC3.** Receive, clarify, and interpret briefs and specifications
- PC4.** Offer and discuss options and alternatives
- PC5.** Discuss time, costs and fees with client, to reach mutual acceptance
- PC6.** Document and sort out customer needs
- PC7.** Use project management skills and techniques to make the most of workplace organization and resources
- PC8.** Follow instructions from available guidance documentation
- PC9.** Record each stage of work development
- PC10.** Keep client regularly updated on progress
- PC11.** Present proposed and final software solutions
- PC12.** Prioritize and schedule tasks
- PC13.** Allocate resources to tasks

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communication and Interpersonal Skills for Mobile Application Development</i>	10	20	-	-
PC1. Prepare for meetings with clients and associates	1	1	-	-
PC2. Gather, clarify, and confirm client requirements	1	1	-	-
PC3. Receive, clarify, and interpret briefs and specifications	1	1	-	-
PC4. Offer and discuss options and alternatives	1	1	-	-
PC5. Discuss time, costs and fees with client, to reach mutual acceptance	1	1	-	-
PC6. Document and sort out customer needs	1	1	-	-
PC7. Use project management skills and techniques to make the most of workplace organization and resources	1	2	-	-
PC8. Follow instructions from available guidance documentation	0.5	2	-	-
PC9. Record each stage of work development	0.5	2	-	-
PC10. Keep client regularly updated on progress	0.5	2	-	-
PC11. Present proposed and final software solutions	0.5	2	-	-
PC12. Prioritize and schedule tasks	0.5	2	-	-
PC13. Allocate resources to tasks	0.5	2	-	-
NOS Total	10	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8438
NOS Name	Communication and Interpersonal Skills for Mobile Application Development
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8439: Sustainable Practice

Description

Sustainable Practice

Scope

The scope covers the following :

- Sustainable Practice

Elements and Performance Criteria

Sustainable Practice

To be competent, the user/individual on the job must be able to:

- PC1.** Select and use tips for writing efficient code, including optimizing algorithms, avoiding double calculations, using memory and resources reasonably, etc.
- PC2.** Apply the principles of sustainable software design, such as modularity, loose coupling, scalability, reusability and accessibility, so that the software can adapt to future changes in requirements
- PC3.** Apply the methods and tools of software testing and quality control to ensure the quality and stability of the software and reduce the cost of repairing bugs and defects
- PC4.** Use version control systems and collaboration tools so team members can develop and manage code collaboratively to improve work efficiency
- PC5.** Practise continuous integration and delivery, and improve software delivery speed and quality through automated testing and deployment processes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sustainable Practice</i>	10	20	-	-
PC1. Select and use tips for writing efficient code, including optimizing algorithms, avoiding double calculations, using memory and resources reasonably, etc.	2	4	-	-
PC2. Apply the principles of sustainable software design, such as modularity, loose coupling, scalability, reusability and accessibility, so that the software can adapt to future changes in requirements	2	4	-	-
PC3. Apply the methods and tools of software testing and quality control to ensure the quality and stability of the software and reduce the cost of repairing bugs and defects	2	4	-	-
PC4. Use version control systems and collaboration tools so team members can develop and manage code collaboratively to improve work efficiency	2	4	-	-
PC5. Practise continuous integration and delivery, and improve software delivery speed and quality through automated testing and deployment processes	2	4	-	-
NOS Total	10	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8439
NOS Name	Sustainable Practice
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8440: Initial Planning, Design, and Test Framework

Description

Initial Planning, Design, and Test Framework

Scope

The scope covers the following :

- Initial Planning, Design, and Test Framework

Elements and Performance Criteria

Initial Planning, Design, and Test Framework

To be competent, the user/individual on the job must be able to:

- PC1.** Choose the most suitable development platforms
- PC2.** Use UI design software such as Adobe XD, Sketch and Sigma
- PC3.** Conduct prototype and visual design on the application user interface (UI)
- PC4.** Use UI application specifications of iOS or Android systems
- PC5.** Produce user experience (UE) documentation for applications
- PC6.** Produce standardized documentation of applications' brand image, following clients' brand guidelines
- PC7.** Plan and design marketing solutions for mobile applications store
- PC8.** Plan test cases
- PC9.** Design specifications for writing test reports

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Initial Planning, Design, and Test Framework</i>	15	20	-	-
PC1. Choose the most suitable development platforms	1	2	-	-
PC2. Use UI design software such as Adobe XD, Sketch and Sigma	1	2	-	-
PC3. Conduct prototype and visual design on the application user interface (UI)	1	2	-	-
PC4. Use UI application specifications of iOS or Android systems	2	2	-	-
PC5. Produce user experience (UE) documentation for applications	2	2	-	-
PC6. Produce standardized documentation of applications' brand image, following clients' brand guidelines	2	2	-	-
PC7. Plan and design marketing solutions for mobile applications store	2	2	-	-
PC8. Plan test cases	2	3	-	-
PC9. Design specifications for writing test reports	2	3	-	-
NOS Total	15	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8440
NOS Name	Initial Planning, Design, and Test Framework
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8441: Systems Architecture Planning

Description

Systems Architecture Planning

Scope

The scope covers the following :

- Systems Architecture Planning

Elements and Performance Criteria

Systems Architecture Planning

To be competent, the user/individual on the job must be able to:

- PC1.** Design the interface based on visual design drafts
- PC2.** Review, select and use open source libraries (such as using Gson OKHttp)
- PC3.** Develop corresponding functions according to the features of different mobile devices
- PC4.** Obtain the mobile terminal device's performance parameters from mobile applications
- PC5.** Implement visualized data statistical analysis and screening in mobile applications
- PC6.** Handle common issues caused by servers, databases, etc

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Systems Architecture Planning</i>	50	100	-	-
PC1. Design the interface based on visual design drafts	5	10	-	-
PC2. Review, select and use open source libraries (such as using Gson OKHttp)	10	10	-	-
PC3. Develop corresponding functions according to the features of different mobile devices	10	20	-	-
PC4. Obtain the mobile terminal device's performance parameters from mobile applications	10	20	-	-
PC5. Implement visualized data statistical analysis and screening in mobile applications	10	20	-	-
PC6. Handle common issues caused by servers, databases, etc	5	20	-	-
NOS Total	50	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8441
NOS Name	Systems Architecture Planning
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8442: Implementation and Product Development

Description

Implementation and Product Development

Scope

The scope covers the following :

- Implementation and Product Development

Elements and Performance Criteria

Implementation and Product Development

To be competent, the user/individual on the job must be able to:

- PC1.** Conduct integrated development with existing code using API (application programming interface)
- PC2.** Realize user interaction effects, animation, and data interaction via programming
- PC3.** Create modular and reusable development codes
- PC4.** Develop Android or iOS interface, and complete compatibility testing
- PC5.** Use Android or iOS development language to implement application development in common design patterns
- PC6.** Use high-performance programming and performance tuning on Android or iOS platform
- PC7.** Apply the test cases, record test results and resolve issues
- PC8.** Plan and implement frequent tests to ensure efficient development
- PC9.** Record test results and resolve issues
- PC10.** Debug the mobile applications to identify issues and write normalized codes to resolve the issues
- PC11.** Complete interface and functional compatibility testing on different platforms and screen resolutions
- PC12.** Simulate testing and troubleshooting of sensors on different devices
- PC13.** Record test results
- PC14.** Implement automated tests of the standardized application programming interfaces
- PC15.** Conduct performance testing and performance tuning (APIs)

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implementation and Product Development</i>	40	60	-	-
PC1. Conduct integrated development with existing code using API (application programming interface)	3	5	-	-
PC2. Realize user interaction effects, animation, and data interaction via programming	3	5	-	-
PC3. Create modular and reusable development codes	3	5	-	-
PC4. Develop Android or iOS interface, and complete compatibility testing	3	5	-	-
PC5. Use Android or iOS development language to implement application development in common design patterns	2	5	-	-
PC6. Use high-performance programming and performance tuning on Android or iOS platform	2	5	-	-
PC7. Apply the test cases, record test results and resolve issues	2	5	-	-
PC8. Plan and implement frequent tests to ensure efficient development	3	5	-	-
PC9. Record test results and resolve issues	3	5	-	-
PC10. Debug the mobile applications to identify issues and write normalized codes to resolve the issues	2	2.5	-	-
PC11. Complete interface and functional compatibility testing on different platforms and screen resolutions	2	2.5	-	-
PC12. Simulate testing and troubleshooting of sensors on different devices	3	2.5	-	-
PC13. Record test results	3	2.5	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Implement automated tests of the standardized application programming interfaces	3	2.5	-	-
PC15. Conduct performance testing and performance tuning (APIs)	3	2.5	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8442
NOS Name	Implementation and Product Development
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8443: Final Product Tests, Troubleshooting, and Optimization

Description

Final Product Tests, Troubleshooting, and Optimization

Scope

The scope covers the following :

- Final Product Tests, Troubleshooting, and Optimization

Elements and Performance Criteria

Final Product Tests, Troubleshooting, and Optimization

To be competent, the user/individual on the job must be able to:

- PC1.** Complete all tests to verify functionality
- PC2.** Bring together all aspects of the project
- PC3.**
 - Analyse and evaluate each stage of the project, relative to
 - • The client's specification
 - • The quality of the user experience
- PC4.** Bring together test results to produce a final report
- PC5.** Evaluate own performance relative to the given brief
- PC6.** . Review the completed project in order to ask "how would I move this on?" "How would I take this to the next stage?"
- PC7.** Prepare and present proposals for optimization to line managers and/or clients

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Final Product Tests, Troubleshooting, and Optimization</i>	40	60	-	-
PC1. Complete all tests to verify functionality	8	10	-	-
PC2. Bring together all aspects of the project	7	10	-	-
PC3. - Analyse and evaluate each stage of the project, relative to - The client's specification - The quality of the user experience	8	10	-	-
PC4. Bring together test results to produce a final report	4	10	-	-
PC5. Evaluate own performance relative to the given brief	4	10	-	-
PC6. . Review the completed project in order to ask "how would I move this on?" "How would I take this to the next stage?"	4	5	-	-
PC7. Prepare and present proposals for optimization to line managers and/or clients	4	5	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8443
NOS Name	Final Product Tests, Troubleshooting, and Optimization
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Carry out assessments under realistic work pressures found in the normal industry workplace (or simulated workplace).
- Ensure that the range of materials, equipment, and tools that learners use are current and of the type routinely found in the normal industry workplace (or simulated workplace) environments.
- Assess that all tasks and functions are completed in a way, and to a timescale that is acceptable in the normal industry workplace.
- Assign workplace (or simulated workplace) responsibilities that enable learners to meet the requirements of the NOS.
- Productivity levels must be checked to ensure that it reflects those that are found in the work situation being replicated.
- Provide situations that allow learners to interact with the range of personnel and contractors found in the normal industry workplace (or simulated workplace).

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8437.Work Organization and Management for Mobile Application Development	20	35	0	0	55	8
SSC/N8438.Communication and Interpersonal Skills for Mobile Application Development	10	20	0	0	30	7
SSC/N8439.Sustainable Practice	10	20	0	0	30	5
SSC/N8440.Initial Planning, Design, and Test Framework	15	20	0	0	35	25
SSC/N8441.Systems Architecture Planning	50	100	0	0	150	15
SSC/N8442.Implementation and Product Development	40	60	0	0	100	30
SSC/N8443.Final Product Tests, Troubleshooting, and Optimization	40	60	0	0	100	10
Total	185	315	-	-	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.